India Employment & Labour Market Performance Report

(Using PLFS Microdata – data.gov.in)

Overview

This report examines India's labour market using PLFS microdata to evaluate employment access, job quality, earnings stability, and structural inequality. The analysis integrates descriptive patterns and statistical validation, including Chi-square testing, to distinguish structural labour market issues from random variation.

Business Problems & Validated Insights

---

1. Youth Unemployment & Age-Based Employment Inequality

Business Question

Are youth (15–29) significantly less likely to be employed than adults?

Statistical Method Applied

- **Chi-square test of independence between age group and employment status**

Key Evidence

- Adults:
  - Observed employed = 69,163
  - Expected employed = 54,289
  - Adults are more employed than expected
- Youth:
  - Observed employed = 17,715
  - Expected employed = 32,589
  - Youth are substantially less employed than expected
- Result: Very high Chi-square value

Insight

Youth employment rates are significantly lower than adult employment rates. The Chi-square test confirms a strong association between age group and employment status, indicating a structural youth employment disadvantage rather than random variation.

---

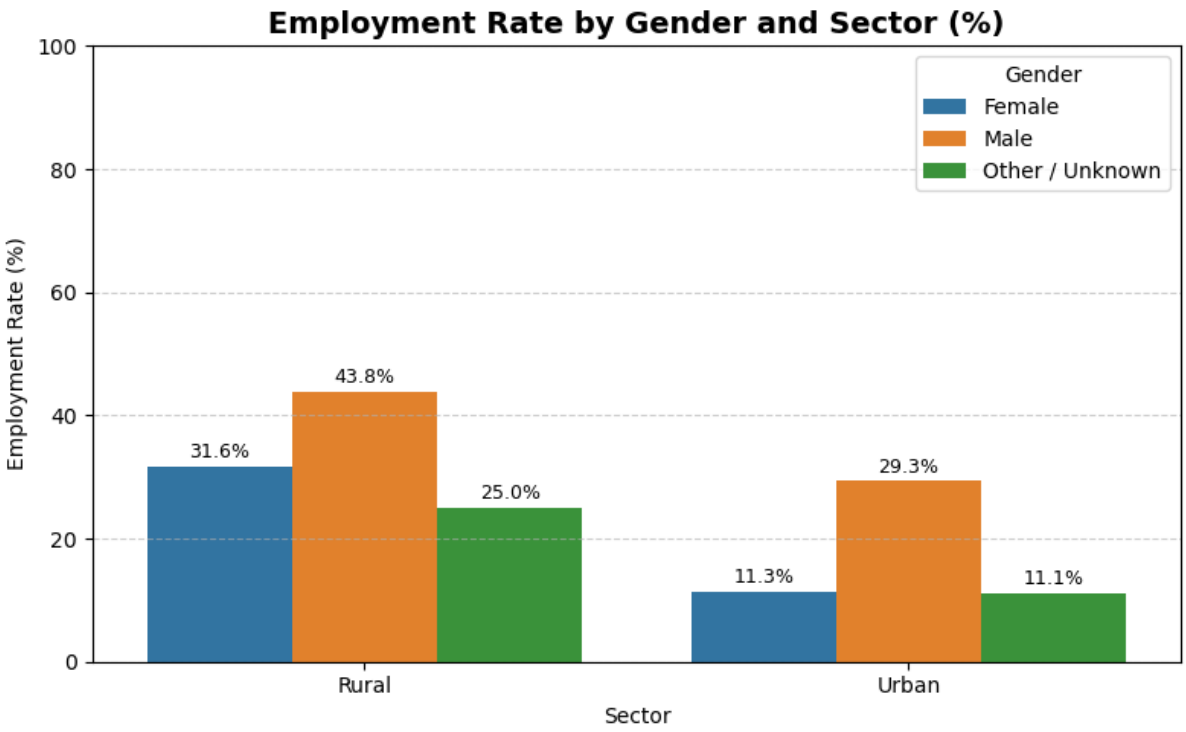
2. Gender Gap in Employment & Earnings

Business Question

Is there a gender-based disparity in employment participation and earnings?

Insight

Men consistently exhibit higher employment rates than women in both rural and urban areas. The gender gap is wider in urban regions, indicating that urban labour markets amplify gender inequality rather than mitigate it. This reflects persistent structural barriers to female labour force participation.



---

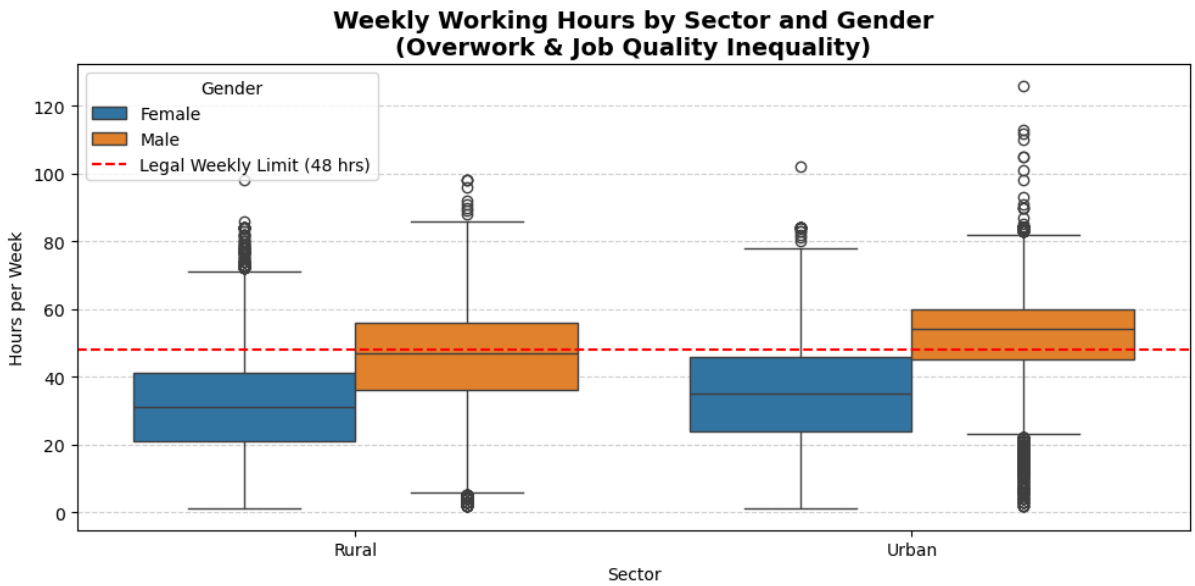
3. Rural vs Urban Employment Quality

Business Question

How does job quality differ between rural and urban employment?

Insight

Urban employment is characterized by longer working hours and a higher risk of exceeding legal working limits, particularly among men, indicating greater work intensity and overwork risk. Rural employment involves fewer working hours but reflects more irregular and unstable job availability. The findings reveal a clear job quality trade-off between higher earning potential with overwork risk in urban areas and lower intensity but fragile employment in rural areas.



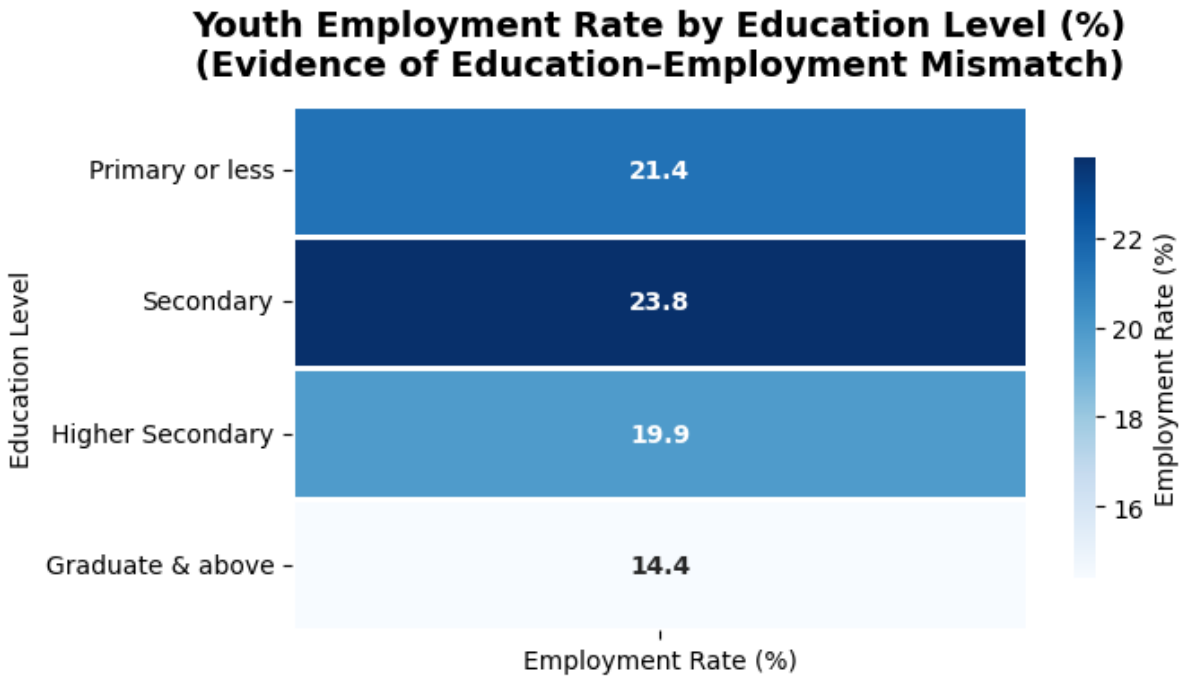
4. Youth Unemployment & Education Mismatch

Business Question

Does higher education translate into better employment outcomes for youth?

Insight

Higher education does not improve employment outcomes for youth. Graduates have the lowest employment rates, while youth with secondary or lower education are more likely to be employed. This highlights a severe education–employment mismatch, driven by misalignment between educational pathways and labour market skill requirements.



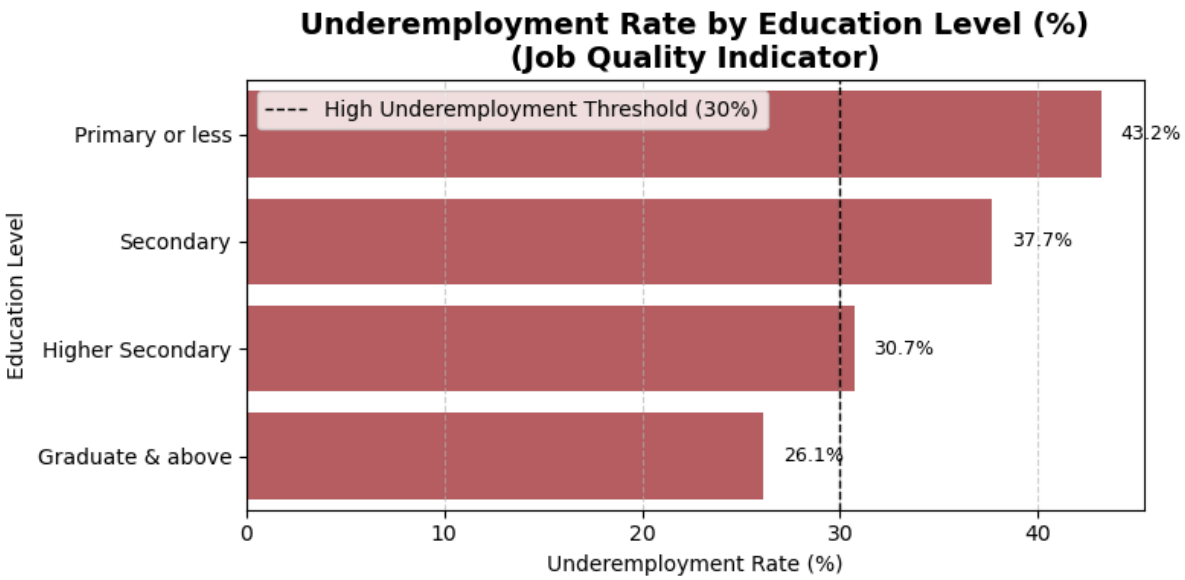
5. Job Quality vs Job Availability (Underemployment)

Business Question

Are employed individuals adequately employed in terms of working hours?

Insight

Underemployment is high across all education levels. Over 40% of employed individuals with lower education are underemployed, reflecting insufficient or irregular working hours. Although underemployment declines with higher education, a notable share of graduates remains underemployed. This confirms a structural issue where job quantity exceeds job quality, and employment alone does not ensure adequate work.



---

6. Earnings Inequality by Employment Type & Gender

Business Question

How do earnings vary across employment types and gender?

Data Reality & Quality Insight

- Self-employed workers report positive income (median ≈ ₹12,000).
- A significant group of employed individuals report zero earnings, resulting in a 100% zero-income rate for this category.
- Salaried income values are largely missing or zero, leading to NaN medians and zero-income rates.

Insight

A substantial share of employed individuals earn no income, indicating income instability and informal work arrangements. Missing salaried income data limits direct wage comparison across employment types. Importantly, NaN values represent lack of valid data, not analytical error, highlighting a critical data-quality constraint alongside real income vulnerability.

---

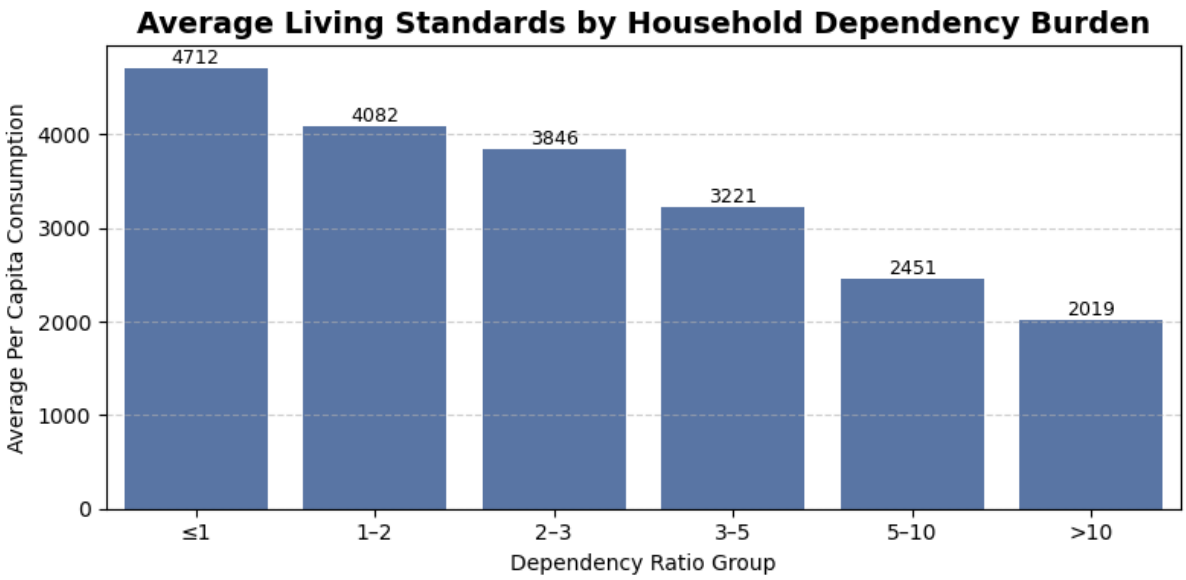
7. Household Economic Condition vs Employment Outcomes

Business Question

How does household earning capacity affect living standards?

Insight

Households with higher dependency burdens experience significantly lower living standards, as limited earning capacity must support more dependents. The findings demonstrate that multiple earners per household, not just individual employment, are essential for economic security.



---

8. Social Group Inequality in Employment & Earnings

Business Question

Do employment and earnings outcomes vary across social groups?

Insight

Employment and earnings outcomes differ sharply across social groups. Some groups (e.g., ST) show higher employment rates but low earnings, while others earn substantially more despite lower employment participation. Socially disadvantaged groups such as SC face a double disadvantage of both lower employment and lower earnings, confirming persistent structural inequality.

	Social group label	Employment rate	Avg earnings
0	OBC	30.85	6356.333426
1	General	29.36	9229.681396
2	SC	23.07	4857.325063
3	ST	38.17	6623.954948

---

9. Seasonal & Quarterly Employment Stability

Business Question

Do employment and earnings fluctuate across calendar quarters?

Insight

Employment and earnings exhibit moderate seasonal variation, peaking in Q1–Q2 and declining in Q4. This pattern indicates seasonal instability in labour demand, particularly affecting informal and vulnerable workers, and increases income insecurity toward later quarters.

	Calendar quarter label	employment rate	Avg earnings
0	Q1	30.62	6986.682212
1	Q2	30.87	7085.532935
2	Q3	30.43	6742.048805
3	Q4	28.62	6771.058167

Overall Conclusion

The analysis confirms that employment alone is not a sufficient indicator of economic well-being in India. Structural disadvantages related to age, gender, education, social group, household composition, and seasonality persist across the labour market. Addressing these challenges requires policy interventions focused on job quality, skill alignment, income stability, and household-level employment security, rather than employment creation alone.